

## Assignment 1: Outline & Annotated Bibliography

Authors X, Y & Z

Research question: How the Brain Machine Interface **can** be exploited to treat illnesses caused by abnormal brain activities?

Sub-questions:

- A. What is a brain-machine interface (BMI)
- B. How does BMI work?
- C. How can BMI be helpful for the treatment of stroke?
- D. How can BMI improve attentional function (cure ADHD)?
- E. How can BMI alleviate the symptoms of Parkinson's disease?
- F. Are there any ethical discussions about BMI?
- G. What are the benefits and limitations of BMI current technology?
- H. What are the implications of BMI technology for the future?

Structure:

- Abstract
  - MAIN
- Introduction
  - A, B, MAIN
- Background knowledge
  - B
- Method & results
  - C, D, E
- Discussion
  - F, G, H
- Conclusion
  - G, H, MAIN
- References

Bibliography:

### **Reference 1:**

Qian, X., Loo, B.R.Y., Castellanos, F.X. et al. Brain-computer-interface-based intervention re-normalizes brain functional network topology in children with attention deficit/hyperactivity disorder. *Transl Psychiatry* 8, 149 (2018). <https://doi.org/10.1038/s41398-018-0213-8>

Motive: Although BCI-based training has shown promise in behavioral improvement in children with ADHD, whether and how these behavioral improvements are supported by functional network changes facilitating brain maturation following the BCI-based treatment in children with ADHD remain largely unknown.

Objective: How do the large-scale brain networks reorganize after the 8-week BCI-based attention treatment in children with ADHD using RS-fMRI imaging?

Conclusion: In conclusion, our study revealed the neural mechanism underlying behavioral improvement following a BCI-based training on children with ADHD.

Argumentation:

1. The BCI-based intervention can help re-normalize brain functional network topology among cognitive networks, which is associated with behavioral improvement and facilitate brain maturation in ADHD children.

Help: It shows the effect of BMI for the treatment of ADHD in children, which is an example for the use of BMI for the treatment of attentional disorders.

### **Reference 2:**

Xie, Y. L., Yang, Y. X., Jiang, H., Duan, X. Y., Gu, L. J., Qing, W., Zhang, B., & Wang, Y. X. (2022). Brain-machine interface-based training for improving upper extremity function after stroke: A meta-analysis of randomized controlled trials. *Frontiers in neuroscience*, 16, 949575. <https://doi.org/10.3389/fnins.2022.949575>

Motive: There is no reliable medical evidence to support the effect of BMI-based training on upper extremity function after stroke.

Objective: How the efficacy and safety of BMI-based training can be used for improving upper extremity function after stroke, as well as potential differences in efficacy of different external devices?

Conclusion: BMI-based training improved upper limb motor function and ADL in post-stroke patients.

Argumentation:

1. The controversy surrounding the effectiveness of BMI-based training for post-stroke upper extremity dysfunction persists due to the different types of experimental clinical studies and the wide variation in the quality of the literature.
2. The efficacy of BMI-based training may also be influenced by other parameters, such as the protocol of the intervention and the severity of the stroke.
3. Existing studies suggest that the mechanism of BMI is mainly related to activation of the cerebral cortex, improvement of functional connectivity, and structural integrity of the brain.

Help: It shows how BMI can be used to improve movement of post-stroke patients, which can be used as an example for the use of BMI as a treatment agent.

### **Reference 3:**

Lebedev, Mikhail. "Brain-machine interfaces: an overview" *Translational Neuroscience*, vol. 5, no. 1, 2014, pp. 99-110. <https://doi.org/10.2478/s13380-014-0212-z>

Motive: BMI technology has experienced rapid development in recent years facilitated by computer and robotics technologies

Objectives: How BMI can be used to treat neurological disabilities?

Conclusion: Although many BMI projects are still at the stage of laboratory research, scientists are confident that BMIs will be entering the clinical world and revolutionizing treatment.

Argumentation:

1. There are almost no efficient treatments for motor and sensory disabilities.
2. BMI holds promise to revolutionize the treatment of severe neurological conditions by establishing direct functional connections between intact brain areas and assistive devices.
3. Furthermore, somatosensory sensation could be restored to such patients using a sensory BMI that connects the somatosensory cortex to prosthetic sensors of touch and position.

Help: Provides a general overview of the brain-machine interface: the development of BMI, how it works and how can it be used to treat illness caused by abnormal brain illness.

#### **Reference 4:**

U. Chaudhary, N. Birbaumer, M.R. Curado, "Brain-Machine Interface (BMI) in paralysis", *Annals of Physical and Rehabilitation Medicine*, Vol. 58, Issue 1, 2015, Pages 9-13, ISSN 1877-0657, <https://doi.org/10.1016/j.rehab.2014.11.002>.

Motive: Since Brain-machine interfaces (BMIs) use brain activity to control external devices, it can facilitate paralyzed patients' interaction with the environment.

Objective: This paper aims to discuss how BMI can be used to treat paralysis

Argumentation:

1. BMI represents a promising strategy to establish communication with paralyzed ALS patients as it does not need muscle engagement for its use.
2. Previous studies demonstrated successful communication with ALS patients using EEG-BMI when patients still showed residual eye control, but patients with complete paralysis were unable to communicate with this.

Conclusion: EEG-BMI training was recently demonstrated in a controlled study to significantly improve motor performance in stroke patients with severe paralysis.

Help: Provides a case where BMI can be used to treat paralysis.

#### **Reference 5:**

J. Dale, S. L. Schmidt, K. Mitchell, D. A. Turner, and W. M. Grill, "Evoked potentials generated by deep brain stimulation for Parkinson's disease," *Brain Stimulation*, vol. 15, no. 5, pp. 1040-1047, 2022. DOI: 10.1016/j.brs.2022.07.048.

Motive: There are limitations when simulating with DBS electrode implantation, causing motor side effects such as anxiety and discomfort. The use of biomarkers linked to neural activity such as evoked potentials helps to improve clinically.

**Objective:** The paper reviews the general features, mechanisms and clinical applications of brain-evoked potentials generated by deep brain stimulation intended for Parkinson's disease.

**Conclusion:** Research on DBS EPs has shown their potential as a feedback signal for adaptive DBS and as a biomarker for DBS parameter selection and electrode positioning. However, further research and technical development are needed to fully characterize their mechanisms and overcome the limitations of current recording systems.

**Argumentation:**

1. DBS evoked potentials (EPs) occur on the millisecond timescale and can be found locally at the site of stimulation and remotely in the cortex.
2. EPs arise from a complex integration of antidromic and orthodromic conduction pathway responses, providing valuable information for understanding the mechanisms and circuits involved in symptom treatment.
3. EPs may serve as biomarkers for improving DBS outcomes and function, such as control signals for DBS programming or adaptive DBS.
4. However, critical gaps in our understanding of the mechanisms of EPs and how to measure and apply these signals in the clinical setting still exist.
5. Technical challenges of recording a highly transient signal at sufficient resolution without interference from stimulation artifact present a barrier to understanding better DBS-induced EPs.

**Help:** shows current landscape of evoked potentials from DBS for Parkinson's disease

### **Reference 6:**

Burwell, S., Sample, M. & Racine, E. Ethical aspects of brain computer interfaces: a scoping review. *BMC Med Ethics* 18, 60 (2017).

<https://doi.org/10.1186/s12910-017-0220-y>

**Motive:** The motive of the paper is to provide a scoping review of the ethical aspects of brain-computer interfaces (BCIs).

**Objective:** The objective of the paper is to identify the key ethical issues associated with BCIs and to provide recommendations for addressing these issues.

**Conclusion:** The paper concludes that there is a need for more empirical research on the ethical aspects of BCIs and for the development of ethical guidelines and standards to guide the development and deployment of this technology. The authors emphasize the importance of considering the ethical implications of BCIs and the need for ongoing research and development in this area to ensure that BCIs are developed and deployed in an ethical and responsible manner.

**Argumentation:**

1. BCIs have the potential to significantly enhance the quality of life for individuals with disabilities or neurological disorders, but there are also ethical concerns related to their use.
2. BCIs raise concerns about privacy, autonomy, and the potential for unintended consequences, such as the potential for brain data to be misused or for BCIs to be used for non-medical purposes.
3. There is a need for ethical guidelines and standards to guide the development and deployment of BCIs, as well as ongoing monitoring and evaluation of their use to ensure that they are developed and deployed in an ethical and responsible manner.
4. BCIs raise important questions about the nature of personal identity, agency, and responsibility, as well as the relationship between the mind and the body.
5. The use of BCIs in military or law enforcement contexts raises particular concerns about their potential use in surveillance or interrogation.
6. There are important ethical questions about the accessibility and distribution of BCIs, including questions about equity, justice, and fairness.

Help: The paper provides a comprehensive overview of the ethical issues surrounding the development and use of brain-computer interfaces (BCIs).